Study notes: Module II: free electron theory and band theory

Puri and Babbar, *Solid State Physics and Electronics*, chapters 5 and 6 (printed pages 151–198). Text extracted by Codex from the page images and checked against them. Figures are redrawn for these notes, not copied from the book; the book figure number is in each caption. Problems are not included.

Section 1: Drude–Lorentz classical theory (Puri and Babbar 5.1)

5.1 Drude–Lorentz classical free electron theory (student notes)

Source: book-sources/bs-3-free-electron-theory.html, Puri and Babbar pp. 151–152. No numbered equations appear in this section except the Wiedemann–Franz ratio, which is shown unnumbered. Figure 5.1 is redrawn.

Before you start

- Electric current is charge moving per unit time; Ohm's law is $V = IR$ (Class 12).
- An electron moving under an electric field gains speed, and collisions reduce it (Class 12, drift velocity).
- Heat conduction and electrical conduction both depend on how freely carriers move.

> **PLUS TWO RECALL: CURRENT, DRIFT AND OHM'S LAW**
>
> **Recall.** The current in a wire is $I = neAv_d$, where n is the carrier density, e the charge, A the area and v_d the drift velocity. Ohm's law says $V = IR$.
>
> **Re-derive.**
>
> 1. Take a wire of length ℓ and area A with field $E = V/\ell$.
> 2. Current density $J = I/A = nev_d$.
> 3. Writing v_d as proportional to E gives $J = \sigma E$ with a constant σ (the conductivity). This is Ohm's law in local form.
>
> **Saved.** $J = \sigma E$. Drude's model gives this σ from the collisions.

1. The model

Drude (1900) pictured a metal as positive ion cores with its valence electrons moving freely among them. The electrons are held in the metal by the attraction of the ion cores, but the potential of the cores is taken as constant, and the repulsion between electrons is neglected. The electrons behave like the molecules of a perfect gas, so the model is called the *free electron gas model*. The book lists three differences between the free electron gas and an ordinary gas: the electrons are negatively charged; their concentration is far higher; and they obey the Pauli exclusion principle.

Because the potential of the ion cores is taken as constant, the potential energy of a conduction electron inside the metal is zero, and its total energy is only kinetic. At the surface the electron faces a potential barrier V_0, the difference between its energy inside and just outside the metal. This barrier keeps the electrons inside (Fig. 5.1). In one dimension the metal is therefore a potential box.

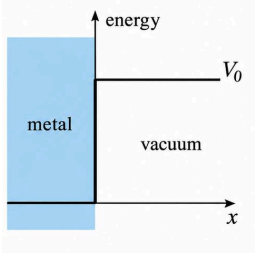


Fig. 5.1 (book): metallic surface bounded by the potential barrier V_0, the difference in potential energy of an electron inside the metal and just outside it.

2. Lorentz's assumption

Drude's picture makes the electrons behave as a perfect gas. Lorentz (1909) assumed that the electrons

obey Maxwell–Boltzmann statistics at equilibrium. The combined theory is the Drude–Lorentz theory. Because it uses classical statistics, it is also called the classical theory.

3. Conduction and Ohm's law

Without a field, the electrons move at random and there is no net current. An applied field accelerates them in one direction. They do not speed up without limit: they collide with the ions and lose their speed. The balance of acceleration and collision gives a steady drift, and the current is proportional to the applied voltage at a fixed temperature. This is Ohm's law. The free motion of the electrons also explains the high electrical and thermal conductivities of metals.

4. Wiedemann–Franz law

Since the electrons move freely whatever the crystal structure, the ratio of electrical to thermal conductivity should be the same for all metals at a given temperature:

$$\frac{\sigma}{\kappa} = \text{constant}$$

The book notes that this law has been verified in practice.

5. Lustre and opacity

Free electrons absorb all incident light and are set into forced oscillation. They re-emit the same energy in all directions, which gives the metallic lustre and the complete opacity of metals.

6. Where the theory fails

- It predicts the room-temperature resistivity correctly, but predicts that resistivity varies as $T^{1/2}$. Experiment shows that it varies linearly with T.
- It fails for the heat capacity and the paramagnetic susceptibility of the conduction electrons. With classical statistics every electron gains energy, which gives values much too large.
- It cannot explain the long mean free paths of electrons at low temperatures (10^8 to 10^9 interatomic spacings, more than a centimetre).

The fixes come from Fermi–Dirac statistics: only a fraction of the electrons can gain energy, and this gives the right heat capacity and susceptibility. That is the content of Sommerfeld's theory (Section 5.2).

> **RESULT: WHAT THE CLASSICAL THEORY GIVES**
>
> Ohm's law, the Wiedemann–Franz ratio independent of metal, lustre and opacity. It fails on the temperature dependence of resistivity, on the heat capacity and on the susceptibility.

Summary

- Metal = fixed ion cores plus a free electron gas inside a potential box of height V_0.
- Drude–Lorentz uses Maxwell–Boltzmann statistics; it explains Ohm's law, the Wiedemann–Franz law and lustre.
- It fails for the heat capacity, susceptibility and the T-dependence of resistivity.

5.2 Sommerfeld's quantum theory: the one-dimensional free electron gas (student notes)

Source: book-sources/bs-3-free-electron-theory.html, Puri and Babbar pp. 153–159. Equations 5.1 to 5.18 are cited by their book numbers. Figures 5.2 to 5.5 are redrawn.

Before you start

- Standing waves on a string of length L fixed at both ends: $L = n\lambda/2$ (Plus Two recall below).
- de Broglie wavelength $\lambda = h/p$ and kinetic energy $E = p^2/2m$.
- Pauli exclusion principle: one electron per quantum state, two states per spatial orbital (spin up and down).

PLUS TWO RECALL: STANDING WAVES ON A STRING WITH FIXED ENDS

Recall. A string of length L with both ends fixed carries standing waves of wavelength $\lambda_n = 2L/n$.

Re-derive.

1. The displacement must vanish at both ends, so the wave is $\sin(n\pi x/L)$ with n a whole number.
2. The wave number is $k = 2\pi/\lambda = n\pi/L$.
3. For a particle of momentum $p = \hbar k$, the kinetic energy is $E = \hbar^2 k^2/2m = \hbar^2 n^2 \pi^2/(2mL^2)$.

Saved. Eqs. (5.8) and (5.10). The electron wave in a box is a standing de Broglie wave.

1. The electron in a one-dimensional box

Take an electron of mass m in a one-dimensional crystal of length L. The walls are treated as infinitely high (Fig. 5.2). Inside, $V = 0$, so the total energy is kinetic:

$$V(x) = 0 \text{ for } 0 < x < L \qquad V(x) = \infty \text{ for } x \le 0 \text{ and } x \ge L \tag{5.1}$$

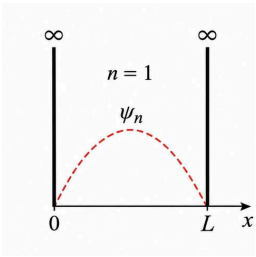


Fig. 5.2 (book): one-dimensional potential box bounded by infinite potential energy barriers.

The wave function ψ_n solves the Schrödinger equation (5.2). Inside the box this becomes (5.3), whose general solution is (5.4), $\psi_n = A\sin kx + B\cos kx$, with $k = \sqrt{2mE_n}/\hbar$ (5.5).

$$\frac{d^2\psi_n}{dx^2} + \frac{2m}{\hbar^2}(E_n - V)\psi_n = 0 \tag{5.2}$$

$$\frac{d^2\psi_n}{dx^2} + \frac{2m}{\hbar^2}E_n\psi_n = 0 \tag{5.3}$$

$$\psi_n(x) = A\sin kx + B\cos kx \tag{5.4}$$

$$k = \sqrt{\frac{2mE_n}{\hbar^2}} \tag{5.5}$$

2. Boundary conditions and the quantum numbers

The wave function must vanish at the walls, because $V\psi_n$ would be infinite there otherwise, and E_n would have to be infinite:

$$\psi_n(0) = 0 \quad \text{and} \quad \psi_n(L) = 0 \tag{5.6}$$

From $\psi_n(0) = 0$, $B = 0$, so $\psi_n = A\sin kx$ (5.7). From $\psi_n(L) = 0$, $\sin kL = 0$, so

$$k = \frac{n\pi}{L}, \qquad n = 1, 2, 3, \ldots \tag{5.8}$$

The wave function is therefore $\psi_n = A\sin(n\pi x/L)$ (5.9), and the energies are

$$E_n = \frac{\hbar^2}{2m}\left(\frac{n\pi}{L}\right)^2 = \frac{n^2h^2}{8mL^2} \quad \text{so} \quad E_n \propto n^2 \tag{5.10}$$

Only integer n give allowed states. The integer n is the quantum number. The levels are spaced by an amount that falls as L grows. For L of a few

centimetres the levels are almost continuous; for atomic L they are widely spaced (Fig. 5.3).

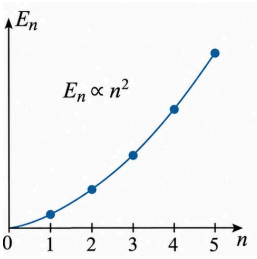


Fig. 5.3 (book): E_n versus n for a one-dimensional crystal.

3. Normalisation

The probability of finding the electron somewhere on the line is one:

$$\int_0^L \psi_n^*(x)\psi_n(x)\,dx = 1$$

Using $\sin^2\theta = \tfrac{1}{2}(1-\cos 2\theta)$, the integral of the cosine term over 0 to L vanishes, leaving $A^2L/2 = 1$, so $A = \sqrt{2/L}$. The normalised wave function is

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi}{L}x\right) \tag{5.11}$$

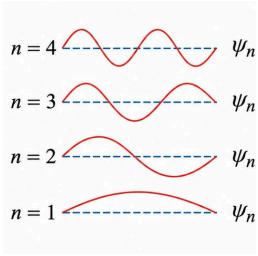


Fig. 5.4 (book): first four wave functions (solid lines) and the corresponding energy levels (broken lines).

4. Filling the levels: the Fermi energy

Electrons fill the levels so that no two share a state. Each level holds two electrons, one of each spin. The top filled level at 0 K is the Fermi level, and its energy is the Fermi energy E_F. For N electrons with the top level n_F:

$$2n_F = N \tag{5.12}$$

Putting $n = n_F$ in (5.10):

$$E_F = \frac{\hbar^2}{2m}\left(\frac{n_F\pi}{L}\right)^2 = \frac{\hbar^2}{2m}\left(\frac{N\pi}{2L}\right)^2 \tag{5.13}$$

E_F depends on the density N/L and on the mass. With $N/L = 0.5$ electrons per Å (5×10^7 per cm), (5.13) gives $E_F = 3.7\times10^{-12}$ erg $= 2.4$ eV.

5. Total energy and average kinetic energy

The total ground-state energy is $E_0 = 2\sum_{n=1}^{N/2} E_n$. Using (5.10) and $\sum_{n=1}^{s} n^2 \simeq \tfrac{1}{3}s^3$ for large s:

$$E_0 = 2\cdot\frac{\hbar^2}{2m}\left(\frac{\pi}{L}\right)^2\sum_{n=1}^{N/2} n^2 \tag{5.14}$$

Substituting the sum and then (5.13) gives $E_0 = \tfrac{1}{3}NE_F$, so the average kinetic energy per electron is one third of E_F (5.15).

$$E_0 = \frac{1}{3}NE_F \tag{5.15}$$

6. Density of states in one dimension

The density of states $D(E)$ is the number of states per unit energy. Each level holds two states, so $D(E) = 2\,dn/dE$ (5.16, 5.17). Using $dE/dn = \hbar^2 n/(4mL^2)$ from (5.10), and then eliminating n with the energy, one obtains

$$\begin{aligned} D(E) &= \left(\frac{8mL^2}{h^2E}\right)^{1/2} \\ &= \frac{4L}{h}\left(\frac{m}{2E}\right)^{1/2} \end{aligned} \tag{5.18}$$

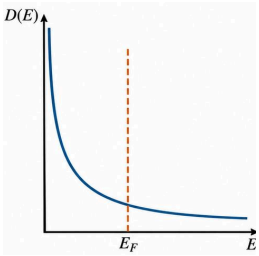


Fig. 5.5 (book): density of states against energy for a one-dimensional metallic crystal; all states up to the Fermi level are filled at 0 K.

All levels below E_F are filled and all above are empty at 0 K (Fig. 5.5).

> **RESULT: 1D FREE ELECTRON GAS**
>
> $E_n = n^2h^2/8mL^2$; $E_F = (\hbar^2/2m)(N\pi/2L)^2$; $E_0 = \tfrac{1}{3}NE_F$; $D(E) \propto E^{-1/2}$.

Summary

- Electrons in a box have quantised energies $E_n \propto n^2$ (Eq. 5.10) and normalised waves (5.11).
- Two electrons per level; the Fermi energy follows from the top level (5.13).
- Average energy $E_0 = \tfrac{1}{3}E_F$ (5.15); density of states $D(E) \propto E^{-1/2}$ (5.18).

Book equations of this section not shown in the derivation above

Listed with their book numbers, as printed (from the verified transcription).

$$\psi_n(x) = A \sin kx \tag{5.7}$$

$$\psi_n(x) = A \sin\left(\frac{n\pi}{L}\right)x \tag{5.9}$$

$$D(E) = \frac{dn}{dE} \tag{5.16}$$

$$D(E) = 2\frac{dn}{dE} \tag{5.17}$$

Section 3: three-dimensional free electron gas and applications (5.2.2, 5.3)

5.2.2–5.3 Free electron gas in three dimensions and its applications (student notes)

Source: book-sources/bs-3-free-electron-theory.html, Puri and Babbar pp. 159–172 (Examples 5.1 and 5.2 included). Equations 5.19 to 5.53 are cited by their book numbers. Figures 5.6 to 5.9 are redrawn. Section 5.3.1 (electronic specific heat) is included.

Before you start

- Volume of a sphere $\frac{4}{3}\pi k^3$ (Plus Two recall below).
- Fermi function: occupation probability of a state at energy E and temperature T.
- Pressure and bulk modulus: $P = -\partial E/\partial V$, $B = -V\,\partial P/\partial V$.

> **PLUS TWO RECALL: VOLUME OF A SPHERE AND COUNTING POINTS**
>
> **Recall.** A sphere of radius k has volume $\frac{4}{3}\pi k^3$.
>
> **Re-derive.**
>
> 1. In k-space each allowed state occupies a cube of side $2\pi/L$, volume $(2\pi/L)^3$.
> 2. The number of k-points inside a sphere of radius k is therefore $\frac{4}{3}\pi k^3/(2\pi/L)^3 = \frac{L^3}{6\pi^2}k^3$.
> 3. Two spin states per point double this to $\frac{L^3}{3\pi^2}k^3$.
>
> **Saved.** The counting in Eq. (5.34).

1. The cube and the wave functions

For a cubical box of edge L, the particle-in-a-box solution is a standing wave in each direction (5.20). The book uses the travelling-wave form $\psi_{\mathbf{k}} = Ae^{i\mathbf{k}\cdot\mathbf{r}}$ (5.21) instead, because it is easier to work with. Periodic boundary conditions over the cube (5.23) fix each component of the wave vector:

$$\psi_{\mathbf{k}}(\mathbf{r}) = Ae^{i\mathbf{k}\cdot\mathbf{r}} \tag{5.21}$$

$$\begin{aligned} k_x &= \frac{2\pi n_x}{L} \\ k_y &= \frac{2\pi n_y}{L} \\ k_z &= \frac{2\pi n_z}{L} \end{aligned} \tag{5.24}$$

The energy of the state is $E_k = \hbar^2k^2/2m = (\hbar^2/2m)(k_x^2 + k_y^2 + k_z^2)$ (5.25), and the normalised wave function is (5.27):

$$\begin{aligned} E_k &= \frac{\hbar^2k^2}{2m} \\ &= \frac{\hbar^2}{2m}\left(k_x^2 + k_y^2 + k_z^2\right) \end{aligned} \tag{5.25}$$

$$\psi_{\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{L^3}}e^{i\mathbf{k}\cdot\mathbf{r}} \tag{5.27}$$

Each state is labelled by k_x, k_y, k_z and the spin m_s. Two spin states share each $\mathbf{k}$, so each energy level holds two electrons.

2. Filling the levels; the Fermi sphere

At 0 K the occupied states form a sphere in k-space (Fig. 5.8). Its surface marks the Fermi wave vector k_F, and the energy at the surface is the Fermi energy (5.33). Counting the states in the sphere (5.34) gives

$$2 \cdot \frac{\frac{4}{3}\pi k_F^3}{(2\pi/L)^3} = \frac{L^3}{3\pi^2} k_F^3 = N \tag{5.34}$$

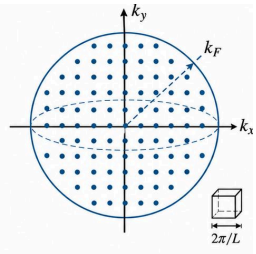


Fig. 5.8 (book): sphere of radius k_F in k-space containing the occupied states of N free electrons in the ground state.

$$k_F = \left(\frac{3\pi^2 N}{V}\right)^{1/3} \tag{5.35}$$

So k_F depends on the density N/V only, not on the mass. Using (5.33) with (5.35):

$$E_F = \frac{\hbar^2}{2m}\left(\frac{3\pi^2 N}{V}\right)^{2/3} \tag{5.36}$$

Inverting (5.36) gives the total number of electrons (5.37), and the Fermi velocity follows from (5.32) and (5.35) (5.38):

$$N = \frac{V}{3\pi^2}\left(\frac{2mE_F}{\hbar^2}\right)^{3/2} \tag{5.37}$$

$$v_F = \frac{\hbar k_F}{m} = \frac{\hbar}{m}\left(\frac{3\pi^2 N}{V}\right)^{1/3} \tag{5.38}$$

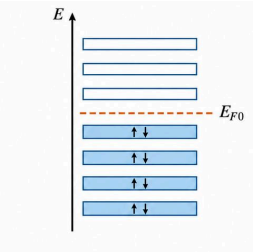


Fig. 5.6 (book): distribution of electrons among the levels of a three-dimensional free electron model at 0 K.

3. Density of states in three dimensions

Require $\int_0^{E_F} D(E)\,dE = N$ (5.39). Using (5.37) and differentiating with respect to E gives the parabolic density of states (5.40):

$$D(E) = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E^{1/2} \tag{5.40}$$

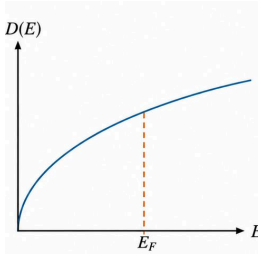


Fig. 5.9 (book): variation of density of electronic states with energy for a free electron gas in three dimensions.

At the Fermi level, $D(E_F) = dN/dE_F$ (5.41). Since $N \propto E_F^{3/2}$, $dN/N = \frac{3}{2}\,dE_F/E_F$ (5.42), so the density of states at E_F is roughly the number of electrons divided by E_F.

$$D(E_F) = \frac{dN}{dE_F} = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E_F^{1/2} \tag{5.41}$$

$$\frac{dN}{N} = \frac{3}{2}\frac{dE_F}{E_F} \tag{5.42}$$

4. The Fermi function and the filled states

At $T > 0$ some electrons leave the levels below E_F, so the occupation is given by the Fermi function (5.28):

$$f(E) = \frac{1}{\exp\left[\dfrac{E - E_F}{k_B T}\right] + 1} \tag{5.28}$$

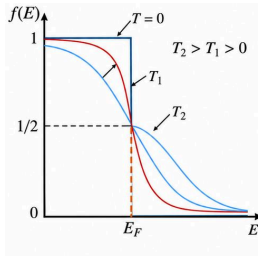


Fig. 5.7 (book): Fermi distribution function at different temperatures; the variation of the Fermi energy with temperature is also shown.

At 0 K it is a step: $f = 1$ below E_{F0} and 0 above (5.29). At $E = E_F$, $f = \frac{1}{2}$ for any T, and the Fermi level moves with temperature as

$$E_F \cong E_{F_0}\left[1 - \frac{\pi^2}{12}\left(\frac{k_B T}{E_{F_0}}\right)^2\right] \tag{5.30}$$

For metals the level spacing is so small that the highest filled level is taken as E_F. Only electrons within about $k_B T$ of E_F are affected by the temperature; at room temperature this is roughly 1% of the electrons, for $E_F = 3$ eV. The filled-state

density is $N(E) = D(E)f(E)$ (5.43), and the number of filled states in dE is $N(E)\,dE$ (5.44).

5. Parameters at absolute zero

(a) **Fermi energy.** Integrating (5.43) with the step function (5.29) gives $N = \int_0^{E_{F0}} D(E)\,dE$; with $n = N/V$:

$$E_{F0} = \frac{\hbar^2}{2m}(3\pi^2 n)^{2/3} \qquad (5.47)$$

(b) **Average kinetic energy.** $E_0 = \frac{1}{N}\int_0^{E_{F0}} E\,D(E)\,dE$. Evaluating the integral with (5.40) and (5.37) gives $E_0 = \frac{3}{5}E_{F0}$ (the average kinetic energy per electron at 0 K). This is the value the classical theory cannot give.

$$E_0 = \tfrac{3}{5}E_{F0}$$

6. Applications (Section 5.3)

The properties of metals divide into static properties (heat capacity, emission, susceptibility, which need only the energy levels) and transport properties (conductivity, Hall effect, which need the response to a field).

6.1 Electronic specific heat

Classical statistics give each electron $\frac{3}{2}k_BT$, so N electrons give $E = \frac{3}{2}Nk_BT$ and $C_V = \frac{3}{2}Nk_B$ (5.49, 5.50). Using one electron per atom, this is about 3 cal g-atom^{-1}K^{-1}, half as large again as the lattice value, which contradicts experiment. The electronic contribution is at most 1% of this at room temperature and falls linearly to zero as $T \to 0$.

Quantum mechanics explains this: only electrons within k_BT of E_F can be excited, which is a fraction $\sim T/T_F$ of the electrons, where $E_F = k_BT_F$. The printed result is

$$C_V \cong \frac{3}{2}Nk_B\left(\frac{T}{T_F}\right) \qquad (5.51)$$

C_V is therefore proportional to T and goes to zero as $T \to 0$. With $T = 300$ K and $T_F = 3000$ K, C_V is about 0.01 of the classical value, in agreement with experiment.

7. Worked examples (book)

Example 5.1. Sodium: $r = 1.86$ Å, bcc, one valence electron, two per unit cell.
$V = a^3 = (4r/\sqrt{3})^3 = 7.93 \times 10^{-29}$ m^3;
$N/V = 2.52 \times 10^{28}$ m^{-3}; $E_{F0} = 4.98 \times 10^{-19}$ J $= 3.11$ eV (from (5.47)).

Example 5.2. From $P = -N\,\partial E_0/\partial V$ with $E_0 = \frac{3}{5}E_{F0}$ and $E_{F0} \propto (N/V)^{2/3}$, the book obtains $P = \frac{2}{5}NE_{F0}/V$ (5.53) and then the pressure–volume law (5.52). The bulk modulus follows from $B = -V\,\partial P/\partial V$ and gives $B = \frac{5}{3}P$, i.e. B in terms of E_0 (the book expresses it through the average kinetic energy).

> **RESULT: THREE-DIMENSIONAL FREE ELECTRON GAS**
>
> $k_F = (3\pi^2 N/V)^{1/3}$; E_F as (5.36); $D(E)$ as (5.40); $E_0 = \frac{3}{5}E_F$; $C_V \cong \frac{3}{2}Nk_B(T/T_F)$.

Summary

- Fermi sphere in k-space; k_F depends only on N/V (5.35).
- E_F (5.36), $D(E) \propto E^{1/2}$ (5.40), $f(E)$ (5.28), $E_F(T)$ (5.30).
- Average kinetic energy $\frac{3}{5}E_{F0}$; specific heat $C_V \propto T$ (5.51).

Book equations of this section not shown in the derivation above

Listed with their book numbers, as printed (from the verified transcription).

$$\nabla^2\psi_k(\mathbf{r}) + \frac{2m}{\hbar^2}E_k\psi_k(\mathbf{r}) = 0 \qquad (5.19)$$

$$\psi_k(\mathbf{r}) = \sqrt{\frac{8}{L^3}}\sin\left(\frac{\pi n_x x}{L}\right)\sin\left(\frac{\pi n_y y}{L}\right)\sin\left(\frac{\pi n_z z}{L}\right) \qquad (5.20)$$

$$k^2 = \frac{2m}{\hbar^2}E_k \qquad (5.22)$$

$$\begin{aligned}\psi(x+L,y,z) &= \psi(x,y,z)\\ \psi(x,y+L,z) &= \psi(x,y,z)\\ \psi(x,y,z+L) &= \psi(x,y,z)\end{aligned} \qquad (5.23)$$

$$k = \frac{2\pi}{\lambda} \qquad (5.26)$$

$$\begin{aligned}f(E) &= 1 \quad \text{for} \quad E \le E_{F0}\\ &= 0 \quad \text{for} \quad E > E_{F0}\end{aligned} \qquad (5.29)$$

$$\mathbf{p}\,\psi_{\mathbf{k}}(\mathbf{r}) = -i\hbar\nabla\psi_{\mathbf{k}}(\mathbf{r}) = \hbar\mathbf{k}\psi_{\mathbf{k}}(\mathbf{r}) \qquad (5.31)$$

6.1–6.3 Band theory: the Bloch theorem and the Kronig–Penney model (student notes)

Source: book-sources/bs-4-band-theory.html, Puri and Babbar pp. 175–186. Equations 6.1 to 6.47 are cited by their book numbers. Figures 6.1 to 6.5 are redrawn; the book figure numbers are in each caption.

Before you start

- A periodic function repeats after a fixed distance a : $V(x+a) = V(x)$.
- Plane waves e^{ikx} and the wave number $k = 2\pi/\lambda$.
- Bragg reflection of waves from a periodic structure (Module I, Section 2.2).

> **PLUS TWO RECALL: BRAGG CONDITION AT NORMAL INCIDENCE**
>
> **Recall.** Waves reflect constructively from periodic planes when the path difference is a whole number of wavelengths.
>
> **Re-derive.**
>
> 1. Put $k = 2\pi/\lambda$, so the condition $n\lambda = 2a$ becomes $k = n\pi/a$.
> 2. These are the wave numbers where the electron wave is reflected by the lattice.
>
> **Saved.** Eqs. (6.41)–(6.42).

6.1 Why a periodic potential is needed

The free electron model treats the potential as constant. It cannot explain why some solids are conductors and others insulators, why the resistivity behaves as it does in insulators, or why semiconductors conduct. The fault is the constant-potential assumption. In a crystal the ion cores are on a periodic lattice, so the potential seen by an electron is periodic with the lattice constant a (Fig. 6.1). The rest of the potential (from the other electrons) is taken as constant.

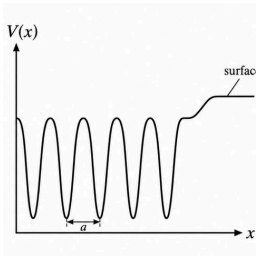


Fig. 6.1 (book): one-dimensional periodic potential experienced by an electron in a crystal with lattice constant a; the potential near the surface is at the right.

Two results follow from a periodic potential: (a) the allowed energies form bands separated by forbidden gaps; (b) $E(\mathbf{k})$ is periodic in $\mathbf{k}$. Neither occurs in the free electron model, where $E = \hbar^2k^2/2m$ has no upper limit.

6.2 The Bloch theorem

For a constant potential V_0 the Schrödinger equation is (6.1) and its solutions are plane waves $e^{\pm ikx}$ with $k^2 = 2m(E - V_0)/\hbar^2$ (6.2). For a periodic potential $V(x) = V(x+a)$, equation (6.3) applies.

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V_0)\psi = 0 \qquad (6.1)$$

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}\big(E - V(x)\big)\psi = 0 \qquad (6.3)$$

Bloch's theorem (Floquet's theorem). The solutions of (6.3) are plane waves modulated by a function with the lattice periodicity:

$$\psi(x) = e^{ikx}u_k(x)$$
$$u_k(x) = u_k(x+a) \qquad (6.4\text{–}6.5)$$

Such functions are the Bloch functions.

Proof outline

Let $g(x)$ and $f(x)$ be two real independent solutions. Because V is periodic, $f(x+a)$ and $g(x+a)$ are also solutions, so each is a combination of $f(x)$ and $g(x)$ (6.7):

$$f(x+a) = \alpha_1 f(x) + \alpha_2 g(x)$$
$$g(x+a) = \beta_1 f(x) + \beta_2 g(x) \qquad (6.7)$$

Choosing combinations A, B that are eigenvectors of this map gives $\psi(x+a)=\lambda\psi(x)$ (6.9–6.10). The eigenvalues satisfy $\lambda^2-(\alpha_1+\beta_2)\lambda+(\alpha_1\beta_2-\alpha_2\beta_1)=0$ (6.12). The Wronskian of f and g is constant, which gives $\alpha_1\beta_2-\alpha_2\beta_1=1$ (6.13), so

$$\lambda^2-(\alpha_1+\beta_2)\lambda+1=0 \tag{6.14}$$

Case I, $(\alpha_1+\beta_2)^2<4$: the roots are complex conjugates $\lambda=e^{\pm ika}$ (6.15), so $\psi(x+a)=e^{ika}\psi(x)$ (6.17). This is the property of a Bloch function, so the theorem follows. These solutions stay bounded.

Case II, $(\alpha_1+\beta_2)^2>4$: the roots are real, $\lambda=e^{\pm\mu a}$ (6.18). These solutions grow without bound and are not acceptable. The energies giving case I are the allowed bands; those giving case II are the forbidden gaps.

In three dimensions, $\psi_{\mathbf{k}}(\mathbf{r})=e^{i\mathbf{k}\cdot\mathbf{r}}u_{\mathbf{k}}(\mathbf{r})$ (6.20). For zero lattice potential, $u_{\mathbf{k}}$ is constant and the wave is a free electron wave.

6.3 The Kronig–Penney model

Replace the periodic potential by a row of square wells of width a (zero potential) separated by barriers of height V_0 and width b (Fig. 6.2). The period is $a+b$. For $E<V_0$ define $\alpha^2=2mE/\hbar^2$ and $\beta^2=2m(V_0-E)/\hbar^2$. The equations are (6.21–6.25):

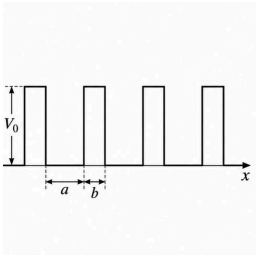


Fig. 6.2 (book): one-dimensional periodic potential used by Kronig and Penney.

$$\begin{aligned}\frac{d^2\psi}{dx^2}+\alpha^2\psi&=0 \quad (0<x<a)\\ \frac{d^2\psi}{dx^2}-\beta^2\psi&=0 \quad (-b<x<0)\end{aligned} \tag{6.23–6.25}$$

The Bloch form $\psi=e^{ikx}u_k(x)$ with u_k periodic in $a+b$ (6.26–6.27) leads to four linear equations in four constants (6.31–6.35). A non-zero solution needs the determinant to vanish, which gives

$$\frac{\beta^2-\alpha^2}{2\alpha\beta}\sinh\beta b\,\sin\alpha a+\cosh\beta b\,\cos\alpha a=\cos k(a+b) \tag{6.36}$$

Delta-function limit. Let $V_0\to\infty$ and $b\to 0$ with V_0b fixed (the barrier strength). Then $\sinh\beta b\to\beta b$, $\cosh\beta b\to 1$, and $(\beta^2+\alpha^2)/2\alpha\beta=mV_0/(\alpha\beta\hbar^2)$. Equation (6.36) becomes (6.37), and with $P=mV_0ba/\hbar^2$ (6.38):

$$\left(\frac{mV_0b}{\hbar^2\alpha}\right)\sin\alpha a+\cos\alpha a=\cos ka \tag{6.37}$$

$$P=\frac{mV_0ba}{\hbar^2} \tag{6.38}$$

$$P\frac{\sin\alpha a}{\alpha a}+\cos\alpha a=\cos ka \tag{6.39}$$

P measures the strength of the barrier: increasing P binds the electron more strongly to each well.

Reading the condition

Since $\cos ka$ lies between -1 and $+1$, only values of αa for which the left side of (6.39) lies between -1 and $+1$ are allowed (Fig. 6.3, $P=3\pi/2$). The conclusions are: (i) the energies form allowed bands separated by forbidden gaps; (ii) the allowed bands widen as αa (energy) grows; (iii) a band narrows as P grows.

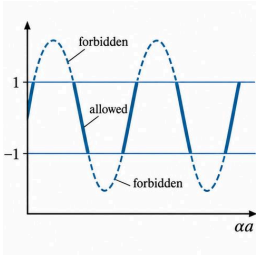


Fig. 6.3 (book): left-hand side of Eq. (6.39) against αa for P = 3π/2; solid and broken lines on the axis are the allowed and forbidden regions.

Limit $P\to\infty$: the allowed bands shrink to lines, which require $\sin\alpha a=0$, i.e. $\alpha a=\pm n\pi$: the levels of a box of atomic size. **Limit $P\to 0$:** $\cos\alpha a=\cos ka$, so $\alpha=k$: the free electron case, with all energies allowed. Figure 6.4 shows the spectrum as a function of P.

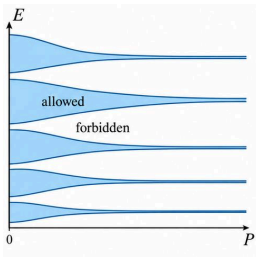


Fig. 6.4 (book): allowed (shaded) and forbidden (open) energy ranges as a function of P.

6.3.1 Three representations of $E(k)$

$\cos ka$ is periodic in k with period $2\pi/a$, so $E(k)$ is too. The periodic (repeated) zone scheme repeats the first zone, $-\pi/a < k < \pi/a$. The extended zone scheme plots each band in its own zone, with discontinuities at $k = \pm n\pi/a$ (6.41). Those values are the zone boundaries, where Bragg reflection occurs. Since $k = 2\pi/\lambda$, (6.41) gives $n\lambda = 2a$ (6.42). The reduced zone scheme brings every point into the first zone by $k = k' - G$ (6.43).

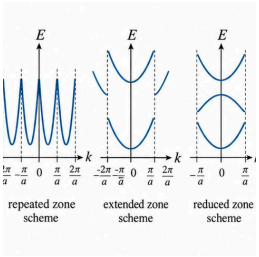


Fig. 6.5 (book): E versus k in three zone schemes: (a) repeated, (b) extended, (c) reduced.

6.3.2 Number of states in a band

For a finite crystal of length L with periodic boundary conditions (6.44), $e^{ikL} = 1$, so $k = 2\pi n/L$ (6.45–6.46) and the number of k-states in dk is $dn = (L/2\pi)dk$ (6.47). The first zone holds N values of k, where N is the number of primitive cells. Each band holds N states and so at most $2N$ electrons (two spins).

> **RESULT: BAND STRUCTURE**
>
> Bloch functions (6.4); allowed bands and gaps (6.15–6.18); Kronig–Penney condition (6.39); each band has N states (6.47).

Summary

- A periodic potential gives bands and gaps; the Bloch theorem (6.4) describes the states.
- Kronig–Penney: allowed bands where $|P \sin \alpha a/\alpha a + \cos \alpha a| \leq 1$ (6.39).
- Each band has N states; $E(k)$ has zone boundaries at $k = n\pi/a$.

Book equations of this section not shown in the derivation above

Listed with their book numbers, as printed (from the verified transcription).

$$\psi(x) = e^{\pm ikx} \tag{6.2}$$

$$\psi(x) = e^{\pm ikx} u_k(x) \tag{6.4}$$

$$u_k(x) = u_k(x + a) \tag{6.5}$$

$$\psi(x) = A\,f(x) + B\,g(x) \tag{6.6}$$

$$\begin{aligned}\psi(x+a) &= A\,f(x+a) + B\,g(x+a) \\ &= (A\alpha_1 + B\beta_1)\,f(x) + (A\alpha_2 + B\beta_2)\,g(x)\end{aligned} \tag{6.8}$$

$$\left.\begin{aligned} A\alpha_1 + B\beta_1 &= \lambda A \\ \text{and} \qquad A\alpha_2 + B\beta_2 &= \lambda B \end{aligned}\right] \tag{6.9}$$

$$\psi(x + a) = \lambda \psi(x) \tag{6.10}$$

$$\begin{vmatrix} (\alpha_1 - \lambda) & \beta_1 \\ \alpha_2 & (\beta_2 - \lambda) \end{vmatrix} = 0 \tag{6.11}$$

$$\lambda^2 - (\alpha_1 + \beta_2)\lambda + \alpha_1\beta_2 - \alpha_2\beta_1 = 0 \tag{6.12}$$

$$\alpha_1\beta_2 - \alpha_2\beta_1 = 1 \tag{6.13}$$

$$\left.\begin{aligned} \lambda_1 &= e^{ika} \\ \text{and} \qquad \lambda_2 &= e^{-ika} \end{aligned}\right] \tag{6.15}$$

$$\left.\begin{aligned} \psi_1(x+a) &= e^{ika}\psi_1(x) \\ \text{and} \qquad \psi_2(x+a) &= e^{-ika}\psi_2(x) \end{aligned}\right] \tag{6.16}$$

$$\psi(x + a) = e^{\pm ika}\,\psi(x) \tag{6.17}$$

$$\lambda_1 = e^{\mu a} \qquad \text{and} \qquad \lambda_2 = e^{-\mu a} \tag{6.18}$$

$$\begin{gathered} \psi_1(x) = e^{\mu x} u(x) \\ \text{and} \\ \psi_2(x) = e^{-\mu x}\,u(x) \end{gathered} \tag{6.19}$$

$$\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}\,u_{\mathbf{k}}(\mathbf{r}) \tag{6.20}$$

$$\begin{gathered} \frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}E\psi = 0 \\ \text{for } 0 < x < a \end{gathered} \tag{6.21}$$

$$\begin{gathered} \frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V_0)\psi = 0 \\ \text{for } -b < x < 0 \end{gathered} \tag{6.22}$$

$$\begin{aligned} \alpha^2 &= \frac{2mE}{\hbar^2} \quad \text{and} \quad \beta^2 \\ &= \frac{2m(V_0 - E)}{\hbar^2} \end{aligned} \tag{6.23}$$

$$\frac{d^2\psi}{dx^2} + \alpha^2\psi = 0 \qquad \text{for } 0 < x < a \tag{6.24}$$

$$\frac{d^2\psi}{dx^2} - \beta^2\psi = 0 \qquad \text{for } -b < x < 0 \tag{6.25}$$

$$\psi(x) = e^{ikx}\,u_k(x) \tag{6.26}$$

$$u_k(x) = u_k(x + a + b) \tag{6.27}$$

$$\frac{d^2\psi}{dx^2} = -k^2 e^{ikx}u_k(x) + 2ike^{ikx}\frac{du_k}{dx} + e^{ikx}\frac{d^2u_k}{dx^2} \tag{6.28}$$

$$\begin{gathered} \frac{d^2u_1}{dx^2} + 2ik\frac{du_1}{dx} + (\alpha^2 - k^2)u_1 = 0 \\ 0 < x < a \end{gathered} \tag{6.29}$$

$$\frac{d^2u_2}{dx^2} + 2ik\frac{du_2}{dx} - (\beta^2 + k^2)u_2 = 0; \quad -b < x < 0 \tag{6.30}$$

$$u_1 = A\,e^{i(\alpha-k)x} + B\,e^{-i(\alpha+k)x}; \qquad 0 < x < a \tag{6.31}$$

$$u_2 = C\,e^{(\beta-ik)x} + D\,e^{-(\beta+ik)x} \quad (-b < x < 0) \tag{6.32}$$

$$u_1(0) = u_2(0); \qquad \left(\frac{du_1}{dx}\right)_{x=0} = \left(\frac{du_2}{dx}\right)_{x=0} \tag{6.33}$$

$$u_1(a) = u_2(-b); \qquad \left(\frac{du_1}{dx}\right)_{x=a} = \left(\frac{du_2}{dx}\right)_{x=-b} \tag{6.34}$$

$$\begin{aligned} A + B &= C + D \\ Ai(\alpha-k) - Bi(\alpha+k) &= C(\beta-ik) - D(\beta+ik) \\ Ae^{i(\alpha-k)a} + Be^{-i(\alpha+k)a} &= Ce^{-(\beta-ik)b} + De^{(\beta+ik)b} \\ Ai(\alpha-k)e^{i(\alpha-k)a} - Bi(\alpha+k)e^{-i(\alpha+k)a} &= C(\beta-ik)e^{-(\beta-ik)b} - D(\beta+ik)e^{(\beta+ik)b} \end{aligned} \tag{6.35}$$

$$E = \frac{\alpha^2\hbar^2}{2m} = \frac{\pi^2\hbar^2}{2ma^2}n^2 \tag{6.40}$$

$$\mathbf{k} = \mathbf{k}' - \mathbf{G} \tag{6.43}$$

$$\psi(x + L) = \psi(x) \tag{6.44}$$

$$k = \frac{2\pi n}{L} \tag{6.45}$$

$$k = \pm\frac{2\pi}{L},\ \pm\frac{4\pi}{L},\ \ldots \tag{6.46}$$

$$dn = \frac{L}{2\pi}dk \tag{6.47}$$

Section 5: velocity, effective mass, metals and semiconductors (6.4–6.5)

6.4–6.5 Velocity and effective mass of an electron; metals, insulators and semiconductors (student notes)

Source: book-sources/bs-4-band-theory.html, Puri and Babbar pp. 189–194 (Examples 6.1 and 6.2 included). Equations 6.48 to 6.59 are cited by their book numbers. Figures 6.6 and 6.8 are redrawn.

Before you start

- Velocity is the rate of change of position; for a wave packet it is the group velocity (box below).
- Force and acceleration: $F = ma$; a field $\mathcal{E}$ on a charge e gives force $e\mathcal{E}$.
- A completely full band has no empty states for an electron to move into.

PLUS TWO RECALL: FORCE, ACCELERATION AND THE CHAIN RULE

Recall. $F = ma$, and the acceleration is the rate of change of velocity.

Re-derive.

1. Write $v = (1/\hbar)\, dE/dk$ (Eq. 6.49).
2. Differentiate with respect to time, using the chain rule: $a = dv/dt = (1/\hbar)(d^2E/dk^2)(dk/dt)$.
3. With $\hbar\, dk/dt = e\mathcal{E}$ (Eq. 6.52), $a = (e\mathcal{E}/\hbar^2)\, d^2E/dk^2$ (Eq. 6.53).

Saved. Eq. (6.53).

6.4.1 Velocity

An electron is a wave packet, and its velocity is the group velocity $v_g = d\omega/dk$ (6.48). With $E = \hbar\omega$:

$$v = \frac{d\omega}{dk} = \frac{1}{\hbar}\frac{dE}{dk} \qquad (6.48\text{–}6.49)$$

For a free electron $E = \hbar^2k^2/2m$ and $v = \hbar k/m = p/m$, which is proportional to k. In a band the slope dE/dk changes. The velocity is zero at $k = 0$ and at the zone edge $k = \pm\pi/a$, where the slope vanishes; it is largest at the inflexion point k_0; beyond it the velocity falls as the energy rises (Fig. 6.6b).

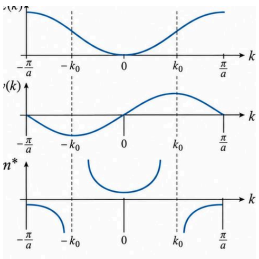


Fig. 6.6 (book): energy, velocity and effective mass as functions of k; dashed lines mark the inflexion points.

6.4.2 Effective mass

With one electron in the zone (so that the Pauli principle is not violated), a field $\mathcal{E}$ acting for dt increases the energy by $dE = e\mathcal{E}v\, dt$. Hence the rate of change of crystal momentum is the force:

$$\hbar\frac{dk}{dt} = \frac{dp}{dt} = F = e\mathcal{E} \qquad (6.52)$$

The acceleration (6.53) differs from the free electron value $a = e\mathcal{E}/m$ (6.54). Equating them defines an effective mass:

$$a = \frac{e\mathcal{E}}{\hbar^2}\frac{d^2E}{dk^2} \qquad (6.53)$$

$$m^* = \frac{\hbar^2}{d^2E/dk^2} \qquad (6.55)$$

m^* depends on the curvature of $E(k)$. It is positive in the lower half of the band, negative in the upper half, and infinite at the inflexion points (Fig. 6.6c). An electron near the top of a band therefore behaves like a positive particle. The book notes that negative mass is explained by Bragg reflection near $k = \pm\pi/a$: a force one way gives momentum the other way.

The factor $f_k = m/m^* = (m/\hbar^2)(d^2E/dk^2)$ (6.56) measures how far the electron behaves as a free one: $f_k = 1$ for a free electron, $f_k < 1$ for a heavy one.

Holes. In a nearly full band, the few empty states near the top behave as particles with positive charge and positive mass, called holes. They carry the conduction in such a band.

6.5 Metals, insulators and semiconductors

Consider a band filled with electrons up to k_1. Its effective number of free electrons is the sum of f_k over occupied states (6.57). With $dn = (L/2\pi)dk$ (6.47) and two electrons per state (6.58):

$$N_{\text{eff}} = \sum_k f_k \qquad (6.57)$$

$$N_{\text{eff}} = \frac{L}{\pi}\int_{-k_1}^{k_1} f_k\, dk \qquad (6.58)$$

Using (6.56), for k_1 within the zone:

$$N_{\text{eff}} = \frac{2Lm}{\pi\hbar^2}\left(\frac{dE}{dk}\right)_{k=k_1} \qquad (6.59)$$

Two results follow. (i) A completely filled band has $N_{\text{eff}} = 0$, because $dE/dk = 0$ at the zone edge. (ii) N_{eff} is largest when dE/dk is largest, i.e. when the band is filled up to the inflexion point.

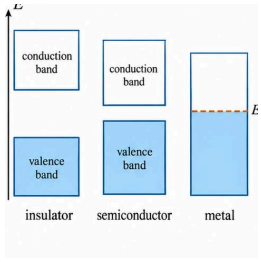


Fig. 6.8 (book): electron distribution at 0 K in (a) an insulator, (b) an intrinsic semiconductor and (c) a metal.

Insulator: complete bands with a wide gap (Fig. 6.8a); no free electrons at any T in practice; diamond has a gap of about 6 eV. **Intrinsic semiconductor**: a narrow gap (about 1 eV, as in germanium and silicon) so that thermal excitation puts some electrons into the conduction band at ordinary T (Fig. 6.8b); all such materials are insulators at 0 K. **Metal**: a partly filled band (Fig. 6.8c); conduction at all T.

The conductivity of semiconductors rises with T because more carriers are excited. The conductivity of metals falls with T, because more phonons scatter the electrons.

Examples (book)

Example 6.1. For $P \ll 1$ in the Kronig–Penney model, the energy of the lowest band (at $k = \pi/a$) is $E \approx \pi^2\hbar^2/2ma^2$ (book, result as printed; the derivation uses (6.39) with $\alpha a \approx P/2$).

Example 6.2. $E = -Ak^2$ with $A = 10^{-39}$ J m^2; at $k = 10^{10}$ m^{-1}: $m_e^* = -\hbar^2/2A = -5.5 \times 10^{-30}$ kg; hole mass $+5.5 \times 10^{-30}$ kg; hole momentum $p_h = -1.05 \times 10^{-24}$ kg m s^{-1}; hole energy 10^{-19} J referred to the top of the band.

> **RESULT: EFFECTIVE MASS AND METAL/INSULATOR TEST**
>
> $v = (1/\hbar)dE/dk$ (6.49); $m^* = \hbar^2/(d^2E/dk^2)$ (6.55); $N_{\text{eff}} = (2Lm/\pi\hbar^2)(dE/dk)_{k_1}$ (6.59), zero for a full band.

Summary

- Velocity (6.49), effective mass (6.55), factor f_k (6.56).
- $N_{\text{eff}} = 0$ for a full band (6.59); metals have partly filled bands, insulators have a wide gap, semiconductors a narrow gap.

Book equations of this section not shown in the derivation above

Listed with their book numbers, as printed (from the verified transcription).

$$v = v_g = \frac{d\omega}{dk} \tag{6.48}$$

$$\begin{aligned} dE &= e\mathcal{E}\,dx \\ &= e\mathcal{E}v\,dt \\ &= \frac{e\mathcal{E}}{\hbar}\left(\frac{dE}{dk}\right)dt \end{aligned} \tag{6.50}$$

$$\frac{dk}{dt} = \frac{e\mathcal{E}}{\hbar} \tag{6.51}$$

$$a = \frac{e\mathcal{E}}{m} \tag{6.54}$$

$$f_k = \frac{m}{m^*} = \frac{m}{\hbar^2}\left(\frac{d^2E}{dk^2}\right) \tag{6.56}$$